

UI/UX Layout: The Command Center

1. Global Navigation & Bilingual Core

- **Top Bar:** EN/ZH Toggle, Global Search (Gene Symbol), and "Export Publication-Ready PDF".
- **Style:** Dark Mode "Lab" Aesthetic (JetBrains Mono font, high-contrast charts).

2. Module 1: The Multi-Omics Atlas

- **Function:** Compare PARL & MAVS across every available database.
- **Options:** Split view (Normal vs. Tumor), Stage filtering (Stage I-IV), and MSI status (Stable vs. Instable).

3. Module 2: The Interaction Lab (PPI)

- **Function:** Visualizing the "Cleavage" mechanism.
- **Visuals:** 3D protein structure viewer (NGLView) and interactive Cytoscape network.
- **Interaction:** User can click an edge to see the supporting PubMed citation.

4. Module 3: Clinical & Prognostic Suite

- **Function:** Predictive modeling for patient survival.
- **Tools:** Kaplan-Meier Plotter with "Optimal Cutpoint" auto-calculation.
- **Feature:** "Synergy Score" — Does high PARL + low MAVS indicate 2x worse survival?

5. Module 4: Drug Discovery (Output)

- **Function:** Recommending therapies.
- **Logic:** Searches for drugs that inhibit PARL or stabilize MAVS.